Origo worksheet

How coin flips, dice and SeedPicker's tickets become a BIP39 seed, one word at a time, and how to check every word by hand. After these pages come a grid to write a run on and the lookup: all 2048 words on four pages, laid out so that your flips or rolls lead straight to the word.

The printed sheet holds no secret: it is the same for everyone. It becomes your seed the moment you write a run on it. From then on keep it like the seed, or destroy it once the words are copied, and never photograph or scan it.

What you need

A pen, the Origo device, and one of:

- Coin flips: any coin.
- D6 dice: one or more ordinary six-sided dice.
- D20 dice: one or more twenty-sided dice.
- SeedPicker tickets: the 342 tickets T1 to T342 from seedpicker.net, printed, cut out and put in a bag or bowl, and one six-sided die. Origo is not affiliated with SeedPicker.

	12 words	24 words
Coin flips	128 flips	256 flips
D6 dice	66 rolls + 7 flips	138 rolls + 3 flips
D20 dice	33 rolls + 7 flips	69 rolls + 3 flips
Tickets	11 draws + 7 flips	23 draws + 3 flips

Rolls are counted as the rolls that count: the dice take more throws than that, as their sections explain.

The idea: eleven bits name a word

The BIP39 list has 2048 words, numbered here and on Origo from 0001 (abandon) to 2048 (zoo). 2048 is 2 to the 11th power, so eleven binary digits - bits, each 0 or 1 - name exactly one word. Every method makes those eleven bits straight from what you flipped, rolled or drew, in order: nothing is hashed and nothing is estimated. A word's number is its eleven bits read as a binary number, plus 1.

Converting bits to a word number

Write the eleven bits under these place values, add up the place values with a 1 under them, then add 1:

place	1024	512	256	128	64	32	16	8	4	2	1
bits	1	1	0	1	0	0	1	0	1	0	0

1024 + 512 + 128 + 16 + 4 = 1684 1684 + 1 = 1685 -> spoon

Back from a number to bits: subtract 1, then go down the place values from 1024, writing 1 and subtracting the place value when it fits, 0 when it does not. 1685 - 1 = 1684: 1024 fits (660 left), 512 fits (148), 256 not, 128 fits (20), 64 not, 32 not, 16 fits (4), 8 not, 4 fits (0), 2 not, 1 not: 11010010100.

The lookup has done this for every word, so a run needs no arithmetic; it is here so that the tables themselves can be checked. Each method below also has its own formula in ordinary numbers.

A run, step by step

1. On Origo choose New Seed, From entropy, the method, then 12 words or 24 words. Origo opens on the method's rule.
2. Write the method and the length at the top of the recording grid.
3. For each word, make its entries - flips, rolls, or a ticket and a die - and write them in that word's row as you enter them on Origo, one at a time. Each entry can be undone.
4. Origo then shows what made the word, its number and the word - 4, 2, 1, 3, 3, 2 and 1685 spoon, say. Find the word in the lookup from what you wrote (see Reading the lookup) and check that it is the same. If it is, write the number and the word in the grid and tick it. If not, decline: that word's entries are thrown away and asked for again.
5. After the last word Origo lists the whole run, four words a page. Compare it with the grid in one pass.
6. Flip the final 7 or 3 coins and write them in the grid's last row. Origo completes the last word (see Finishing) and shows the mnemonic.
7. Copy the mnemonic to where it will be kept, and check the copy against Origo.

Coin flips

Eleven flips make a word. Heads is 1, tails is 0, written in the order flipped: the first flip is the leftmost bit, place value 1024.

```
flips  H H T H   T T H T   H T T
bits   1 1 0 1   0 0 1 0   1 0 0     ->  11010010100  ->  1685  spoon
```

In the lookup the first 8 flips pick the row, under Coins, and the last 3 the column: row 1101 0010, column 100.

D6 dice

Only faces 1 to 4 count. Roll a 5 or a 6 again before entering anything: counting them would make some words likelier than others, and Origo does not offer them. Each counted face gives two bits, the face minus 1 in binary:

```
face   1    2    3    4
bits   00   01   10   11
```

A word is six rolls. The first five give two bits each; the sixth gives only one bit, the first of its two, so on the last roll 1 or 2 means 0 and 3 or 4 means 1.

```
rolls  4    2    1    3    3    2
bits   11   01   00   10   10   0     ->  11010010100  ->  1685  spoon
```

Without bits: number = (1st - 1) × 512 + (2nd - 1) × 128 + (3rd - 1) × 32 + (4th - 1) × 8 + (5th - 1) × 2 + (0 if the 6th is 1 or 2, 1 if it is 3 or 4) + 1. Here 1536 + 128 + 0 + 16 + 4 + 0 + 1 = 1685.

In the lookup the first four rolls pick the row, under D6, and the fifth and sixth the column: row 4 2 1 3, column 3 1-2, where 1-2 means that either face makes the word.

D20 dice

Only faces 1 to 16 count; roll 17, 18, 19 or 20 again. Each counted face gives four bits, the face minus 1 in binary:

```
face   1    2    3    4    5    6    7    8
bits   0000 0001 0010 0011 0100 0101 0110 0111
face   9    10   11   12   13   14   15   16
bits   1000 1001 1010 1011 1100 1101 1110 1111
```

A word is three rolls. The first two give four bits each; the third gives only its first three, so on the last roll the faces pair up: 1 or 2, 3 or 4, and so on to 15 or 16.

```
rolls  16     1      10
bits   1111   0000   100             ->  11110000100  ->  1925  vacuum
```

The last roll, 10, is 1001; its first three bits, 100, are kept. Without bits: number = (1st - 1) × 128 + (2nd - 1) × 8 + half of (3rd - 1), rounded down, + 1. Here 1920 + 0 + 4 + 1 = 1925.

In the lookup the first two rolls pick the row, under D20, and the third the column: row 16 1, column 9-10.

SeedPicker tickets

For each word, draw a ticket from the bag, note it, put it back and mix; then roll the six-sided die, on which every face counts. The pair names word number (ticket - 1) × 6 + die.

```
T57, die 4   ->  (57 - 1) x 6 + 4 = 340   ->  0340  clerk
```

342 tickets and 6 faces make 2052 pairs: the 2048 words once each, and four more, T342 with a 3, 4, 5 or 6, which are past 2048. On those, draw a new ticket - not just a new die, which would make T342's two words three times as likely as any other. Origo says so when it happens.

Find the number in the lookup: each row holds eight words, from the number under No. onwards, so 0340 is the fourth word of row 0337. SeedPicker's own printed table also gives each pair's word directly.

Reading the lookup

The lookup holds the 2048 words in 256 rows of eight, in number order, one D6 first roll to a page. Your first entries pick the row, read in the columns at its left; your last entries pick one of its eight words, read in the three heading lines at the top. Rows are banded in fours, one D6 fourth roll each, and ruled every sixteen, one D20 first roll each.

```
Coins  1101 0010 | 100     row 1101 0010, column 100     ->  spoon
D6     4 2 1 3 | 3 2       row 4 2 1 3, column 3 1-2     ->  spoon
D20    16 1 | 10           row 16 1, column 9-10         ->  vacuum
```

A word's number is its row's number plus its place in the row, counting from 0: spoon is in row 1681, place 4, so 1685.

Finishing: the last word

A 12-word seed is 128 bits of yours and a 4-bit checksum, 132 = 12 × 11. Eleven words make 121 bits, so 7 coin flips finish your 128. For 24 words, 23 words make 253 bits, 3 flips finish 256, and the checksum is 8 bits. Origo then computes the checksum, from SHA-256 of your bits, which no table can do, and with it the last word.

Part of the last word can still be checked here: its first bits are your final flips, 7 of them for 12 words and 3 for 24. Seven flips H T T H H T H are 1001101, so the twelfth word is in one of the two rows whose Coins begin 1001 101, numbers 1233 to 1248. The checksum itself can be checked on an offline computer with tools/origo_verify.py.

SeedPicker's calculator always takes the alphabetically first valid 24th word: three tails give that same word.

Words already chosen

Words drawn elsewhere - SeedPicker's own form, a metal plate, any other table - are finished with New Seed, Complete checksum, then 11 words + 7 coins or 23 words + 3 coins. Type each word, or choose Enter word numbers and type its number from the lookup; then flip the coins as above.

Good practice

- Work alone, with no phone or camera that can see the sheet, the dice or the screen.
- Roll dice from a cup onto a flat, hard surface, and flip coins high enough to turn several times. Never choose a face or a ticket: every entry must come from the throw or the draw.
- Write each entry down before entering it, so the sheet and the device can be compared.
- Use one blank grid per seed, and never reuse a run.
- When the mnemonic is safely copied, keep the filled sheet with it or destroy it - burnt, or cross-cut shredded - and never leave it in a bin whole.

Record a run

One row per word. Tick the last column once the lookup gives the same word.

Method _____ Words: 12 / 24

Word	Rolls, flips or draw	No.	Word	Checked
1				
2				
3				
4				
5				
6				
7				
8				
9				
10				
11				
12				
13				
14				
15				
16				
17				
18				
19				
20				
21				
22				
23				
24				

Final coins (7 or 3)

Lookup: words 0001 to 0512

D6 first roll 1, D20 first roll 1 to 4, coins begin 00. Find the row by your first entries and the column by your last.

					Coins: last 3 flips	000	001	010	011	100	101	110	111
					D6: 5th roll, 6th roll	1 1-2	1 3-4	2 1-2	2 3-4	3 1-2	3 3-4	4 1-2	4 3-4
					D20: 3rd roll	1-2	3-4	5-6	7-8	9-10	11-12	13-14	15-16
No.	**Coins**		**D6**		**D20**								
0001	0000	0000	1 1 1 1	1	1	abandon	ability	able	about	above	absent	absorb	abstract
0009	0000	0001	1 1 1 2	1	2	absurd	abuse	access	accident	account	accuse	achieve	acid
0017	0000	0010	1 1 1 3	1	3	acoustic	acquire	across	act	action	actor	actress	actual
0025	0000	0011	1 1 1 4	1	4	adapt	add	addict	address	adjust	admit	adult	advance
0033	0000	0100	1 1 2 1	1	5	advice	aerobic	affair	afford	afraid	again	age	agent
0041	0000	0101	1 1 2 2	1	6	agree	ahead	aim	air	airport	aisle	alarm	album
0049	0000	0110	1 1 2 3	1	7	alcohol	alert	alien	all	alley	allow	almost	alone
0057	0000	0111	1 1 2 4	1	8	alpha	already	also	alter	always	amateur	amazing	among
0065	0000	1000	1 1 3 1	1	9	amount	amused	analyst	anchor	ancient	anger	angle	angry
0073	0000	1001	1 1 3 2	1	10	animal	ankle	announce	annual	another	answer	antenna	antique
0081	0000	1010	1 1 3 3	1	11	anxiety	any	apart	apology	appear	apple	approve	april
0089	0000	1011	1 1 3 4	1	12	arch	arctic	area	arena	argue	arm	armed	armor
0097	0000	1100	1 1 4 1	1	13	army	around	arrange	arrest	arrive	arrow	art	artefact
0105	0000	1101	1 1 4 2	1	14	artist	artwork	ask	aspect	assault	asset	assist	assume
0113	0000	1110	1 1 4 3	1	15	asthma	athlete	atom	attack	attend	attitude	attract	auction
0121	0000	1111	1 1 4 4	1	16	audit	august	aunt	author	auto	autumn	average	avocado
0129	0001	0000	1 2 1 1	2	1	avoid	awake	aware	away	awesome	awful	awkward	axis
0137	0001	0001	1 2 1 2	2	2	baby	bachelor	bacon	badge	bag	balance	balcony	ball
0145	0001	0010	1 2 1 3	2	3	bamboo	banana	banner	bar	barely	bargain	barrel	base
0153	0001	0011	1 2 1 4	2	4	basic	basket	battle	beach	bean	beauty	because	become
0161	0001	0100	1 2 2 1	2	5	beef	before	begin	behave	behind	believe	below	belt
0169	0001	0101	1 2 2 2	2	6	bench	benefit	best	betray	better	between	beyond	bicycle
0177	0001	0110	1 2 2 3	2	7	bid	bike	bind	biology	bird	birth	bitter	black
0185	0001	0111	1 2 2 4	2	8	blade	blame	blanket	blast	bleak	bless	blind	blood
0193	0001	1000	1 2 3 1	2	9	blossom	blouse	blue	blur	blush	board	boat	body
0201	0001	1001	1 2 3 2	2	10	boil	bomb	bone	bonus	book	boost	border	boring
0209	0001	1010	1 2 3 3	2	11	borrow	boss	bottom	bounce	box	boy	bracket	brain
0217	0001	1011	1 2 3 4	2	12	brand	brass	brave	bread	breeze	brick	bridge	brief
0225	0001	1100	1 2 4 1	2	13	bright	bring	brisk	broccoli	broken	bronze	broom	brother
0233	0001	1101	1 2 4 2	2	14	brown	brush	bubble	buddy	budget	buffalo	build	bulb
0241	0001	1110	1 2 4 3	2	15	bulk	bullet	bundle	bunker	burden	burger	burst	bus
0249	0001	1111	1 2 4 4	2	16	business	busy	butter	buyer	buzz	cabbage	cabin	cable
0257	0010	0000	1 3 1 1	3	1	cactus	cage	cake	call	calm	camera	camp	can
0265	0010	0001	1 3 1 2	3	2	canal	cancel	candy	cannon	canoe	canvas	canyon	capable
0273	0010	0010	1 3 1 3	3	3	capital	captain	car	carbon	card	cargo	carpet	carry
0281	0010	0011	1 3 1 4	3	4	cart	case	cash	casino	castle	casual	cat	catalog
0289	0010	0100	1 3 2 1	3	5	catch	category	cattle	caught	cause	caution	cave	ceiling
0297	0010	0101	1 3 2 2	3	6	celery	cement	census	century	cereal	certain	chair	chalk
0305	0010	0110	1 3 2 3	3	7	champion	change	chaos	chapter	charge	chase	chat	cheap
0313	0010	0111	1 3 2 4	3	8	check	cheese	chef	cherry	chest	chicken	chief	child
0321	0010	1000	1 3 3 1	3	9	chimney	choice	choose	chronic	chuckle	chunk	churn	cigar
0329	0010	1001	1 3 3 2	3	10	cinnamon	circle	citizen	city	civil	claim	clap	clarify
0337	0010	1010	1 3 3 3	3	11	claw	clay	clean	clerk	clever	click	client	cliff
0345	0010	1011	1 3 3 4	3	12	climb	clinic	clip	clock	clog	close	cloth	cloud
0353	0010	1100	1 3 4 1	3	13	clown	club	clump	cluster	clutch	coach	coast	coconut
0361	0010	1101	1 3 4 2	3	14	code	coffee	coil	coin	collect	color	column	combine
0369	0010	1110	1 3 4 3	3	15	come	comfort	comic	common	company	concert	conduct	confirm
0377	0010	1111	1 3 4 4	3	16	congress	connect	consider	control	convince	cook	cool	copper
0385	0011	0000	1 4 1 1	4	1	copy	coral	core	corn	correct	cost	cotton	couch
0393	0011	0001	1 4 1 2	4	2	country	couple	course	cousin	cover	coyote	crack	cradle
0401	0011	0010	1 4 1 3	4	3	craft	cram	crane	crash	crater	crawl	crazy	cream
0409	0011	0011	1 4 1 4	4	4	credit	creek	crew	cricket	crime	crisp	critic	crop
0417	0011	0100	1 4 2 1	4	5	cross	crouch	crowd	crucial	cruel	cruise	crumble	crunch
0425	0011	0101	1 4 2 2	4	6	crush	cry	crystal	cube	culture	cup	cupboard	curious
0433	0011	0110	1 4 2 3	4	7	current	curtain	curve	cushion	custom	cute	cycle	dad
0441	0011	0111	1 4 2 4	4	8	damage	damp	dance	danger	daring	dash	daughter	dawn
0449	0011	1000	1 4 3 1	4	9	day	deal	debate	debris	decade	december	decide	decline
0457	0011	1001	1 4 3 2	4	10	decorate	decrease	deer	defense	define	defy	degree	delay
0465	0011	1010	1 4 3 3	4	11	deliver	demand	demise	denial	dentist	deny	depart	depend
0473	0011	1011	1 4 3 4	4	12	deposit	depth	deputy	derive	describe	desert	design	desk
0481	0011	1100	1 4 4 1	4	13	despair	destroy	detail	detect	develop	device	devote	diagram
0489	0011	1101	1 4 4 2	4	14	dial	diamond	diary	dice	diesel	diet	differ	digital
0497	0011	1110	1 4 4 3	4	15	dignity	dilemma	dinner	dinosaur	direct	dirt	disagree	discover
0505	0011	1111	1 4 4 4	4	16	disease	dish	dismiss	disorder	display	distance	divert	divide

Lookup: words 0513 to 1024

D6 first roll 2, D20 first roll 5 to 8, coins begin 01. Find the row by your first entries and the column by your last.

					Coins: last 3 flips	000	001	010	011	100	101	110	111
					D6: 5th roll, 6th roll	1 1-2	1 3-4	2 1-2	2 3-4	3 1-2	3 3-4	4 1-2	4 3-4
					D20: 3rd roll	1-2	3-4	5-6	7-8	9-10	11-12	13-14	15-16
No.	**Coins**	**D6**	**D20**										
0513	0100 0000	2 1 1 1	5	1		divorce	dizzy	doctor	document	dog	doll	dolphin	domain
0521	0100 0001	2 1 1 2	5	2		donate	donkey	donor	door	dose	double	dove	draft
0529	0100 0010	2 1 1 3	5	3		dragon	drama	drastic	draw	dream	dress	drift	drill
0537	0100 0011	2 1 1 4	5	4		drink	drip	drive	drop	drum	dry	duck	dumb
0545	0100 0100	2 1 2 1	5	5		dune	during	dust	dutch	duty	dwarf	dynamic	eager
0553	0100 0101	2 1 2 2	5	6		eagle	early	earn	earth	easily	east	easy	echo
0561	0100 0110	2 1 2 3	5	7		ecology	economy	edge	edit	educate	effort	egg	eight
0569	0100 0111	2 1 2 4	5	8		either	elbow	elder	electric	elegant	element	elephant	elevator
0577	0100 1000	2 1 3 1	5	9		elite	else	embark	embody	embrace	emerge	emotion	employ
0585	0100 1001	2 1 3 2	5	10		empower	empty	enable	enact	end	endless	endorse	enemy
0593	0100 1010	2 1 3 3	5	11		energy	enforce	engage	engine	enhance	enjoy	enlist	enough
0601	0100 1011	2 1 3 4	5	12		enrich	enroll	ensure	enter	entire	entry	envelope	episode
0609	0100 1100	2 1 4 1	5	13		equal	equip	era	erase	erode	erosion	error	erupt
0617	0100 1101	2 1 4 2	5	14		escape	essay	essence	estate	eternal	ethics	evidence	evil
0625	0100 1110	2 1 4 3	5	15		evoke	evolve	exact	example	excess	exchange	excite	exclude
0633	0100 1111	2 1 4 4	5	16		excuse	execute	exercise	exhaust	exhibit	exile	exist	exit
0641	0101 0000	2 2 1 1	6	1		exotic	expand	expect	expire	explain	expose	express	extend
0649	0101 0001	2 2 1 2	6	2		extra	eye	eyebrow	fabric	face	faculty	fade	faint
0657	0101 0010	2 2 1 3	6	3		faith	fall	false	fame	family	famous	fan	fancy
0665	0101 0011	2 2 1 4	6	4		fantasy	farm	fashion	fat	fatal	father	fatigue	fault
0673	0101 0100	2 2 2 1	6	5		favorite	feature	february	federal	fee	feed	feel	female
0681	0101 0101	2 2 2 2	6	6		fence	festival	fetch	fever	few	fiber	fiction	field
0689	0101 0110	2 2 2 3	6	7		figure	file	film	filter	final	find	fine	finger
0697	0101 0111	2 2 2 4	6	8		finish	fire	firm	first	fiscal	fish	fit	fitness
0705	0101 1000	2 2 3 1	6	9		fix	flag	flame	flash	flat	flavor	flee	flight
0713	0101 1001	2 2 3 2	6	10		flip	float	flock	floor	flower	fluid	flush	fly
0721	0101 1010	2 2 3 3	6	11		foam	focus	fog	foil	fold	follow	food	foot
0729	0101 1011	2 2 3 4	6	12		force	forest	forget	fork	fortune	forum	forward	fossil
0737	0101 1100	2 2 4 1	6	13		foster	found	fox	fragile	frame	frequent	fresh	friend
0745	0101 1101	2 2 4 2	6	14		fringe	frog	front	frost	frown	frozen	fruit	fuel
0753	0101 1110	2 2 4 3	6	15		fun	funny	furnace	fury	future	gadget	gain	galaxy
0761	0101 1111	2 2 4 4	6	16		gallery	game	gap	garage	garbage	garden	garlic	garment
0769	0110 0000	2 3 1 1	7	1		gas	gasp	gate	gather	gauge	gaze	general	genius
0777	0110 0001	2 3 1 2	7	2		genre	gentle	genuine	gesture	ghost	giant	gift	giggle
0785	0110 0010	2 3 1 3	7	3		ginger	giraffe	girl	give	glad	glance	glare	glass
0793	0110 0011	2 3 1 4	7	4		glide	glimpse	globe	gloom	glory	glove	glow	glue
0801	0110 0100	2 3 2 1	7	5		goat	goddess	gold	good	goose	gorilla	gospel	gossip
0809	0110 0101	2 3 2 2	7	6		govern	gown	grab	grace	grain	grant	grape	grass
0817	0110 0110	2 3 2 3	7	7		gravity	great	green	grid	grief	grit	grocery	group
0825	0110 0111	2 3 2 4	7	8		grow	grunt	guard	guess	guide	guilt	guitar	gun
0833	0110 1000	2 3 3 1	7	9		gym	habit	hair	half	hammer	hamster	hand	happy
0841	0110 1001	2 3 3 2	7	10		harbor	hard	harsh	harvest	hat	have	hawk	hazard
0849	0110 1010	2 3 3 3	7	11		head	health	heart	heavy	hedgehog	height	hello	helmet
0857	0110 1011	2 3 3 4	7	12		help	hen	hero	hidden	high	hill	hint	hip
0865	0110 1100	2 3 4 1	7	13		hire	history	hobby	hockey	hold	hole	holiday	hollow
0873	0110 1101	2 3 4 2	7	14		home	honey	hood	hope	horn	horror	horse	hospital
0881	0110 1110	2 3 4 3	7	15		host	hotel	hour	hover	hub	huge	human	humble
0889	0110 1111	2 3 4 4	7	16		humor	hundred	hungry	hunt	hurdle	hurry	hurt	husband
0897	0111 0000	2 4 1 1	8	1		hybrid	ice	icon	idea	identify	idle	ignore	ill
0905	0111 0001	2 4 1 2	8	2		illegal	illness	image	imitate	immense	immune	impact	impose
0913	0111 0010	2 4 1 3	8	3		improve	impulse	inch	include	income	increase	index	indicate
0921	0111 0011	2 4 1 4	8	4		indoor	industry	infant	inflict	inform	inhale	inherit	initial
0929	0111 0100	2 4 2 1	8	5		inject	injury	inmate	inner	innocent	input	inquiry	insane
0937	0111 0101	2 4 2 2	8	6		insect	inside	inspire	install	intact	interest	into	invest
0945	0111 0110	2 4 2 3	8	7		invite	involve	iron	island	isolate	issue	item	ivory
0953	0111 0111	2 4 2 4	8	8		jacket	jaguar	jar	jazz	jealous	jeans	jelly	jewel
0961	0111 1000	2 4 3 1	8	9		job	join	joke	journey	joy	judge	juice	jump
0969	0111 1001	2 4 3 2	8	10		jungle	junior	junk	just	kangaroo	keen	keep	ketchup
0977	0111 1010	2 4 3 3	8	11		key	kick	kid	kidney	kind	kingdom	kiss	kit
0985	0111 1011	2 4 3 4	8	12		kitchen	kite	kitten	kiwi	knee	knife	knock	know
0993	0111 1100	2 4 4 1	8	13		lab	label	labor	ladder	lady	lake	lamp	language
1001	0111 1101	2 4 4 2	8	14		laptop	large	later	latin	laugh	laundry	lava	law
1009	0111 1110	2 4 4 3	8	15		lawn	lawsuit	layer	lazy	leader	leaf	learn	leave
1017	0111 1111	2 4 4 4	8	16		lecture	left	leg	legal	legend	leisure	lemon	lend

Lookup: words 1025 to 1536

D6 first roll 3, D20 first roll 9 to 12, coins begin 10. Find the row by your first entries and the column by your last.

					Coins: last 3 flips	000	001	010	011	100	101	110	111
					D6: 5th roll, 6th roll	1 1-2	1 3-4	2 1-2	2 3-4	3 1-2	3 3-4	4 1-2	4 3-4
					D20: 3rd roll	1-2	3-4	5-6	7-8	9-10	11-12	13-14	15-16
No.	**Coins**	**D6**	**D20**										
1025	1000 0000	3 1 1 1	9	1		length	lens	leopard	lesson	letter	level	liar	liberty
1033	1000 0001	3 1 1 2	9	2		library	license	life	lift	light	like	limb	limit
1041	1000 0010	3 1 1 3	9	3		link	lion	liquid	list	little	live	lizard	load
1049	1000 0011	3 1 1 4	9	4		loan	lobster	local	lock	logic	lonely	long	loop
1057	1000 0100	3 1 2 1	9	5		lottery	loud	lounge	love	loyal	lucky	luggage	lumber
1065	1000 0101	3 1 2 2	9	6		lunar	lunch	luxury	lyrics	machine	mad	magic	magnet
1073	1000 0110	3 1 2 3	9	7		maid	mail	main	major	make	mammal	man	manage
1081	1000 0111	3 1 2 4	9	8		mandate	mango	mansion	manual	maple	marble	march	margin
1089	1000 1000	3 1 3 1	9	9		marine	market	marriage	mask	mass	master	match	material
1097	1000 1001	3 1 3 2	9	10		math	matrix	matter	maximum	maze	meadow	mean	measure
1105	1000 1010	3 1 3 3	9	11		meat	mechanic	medal	media	melody	melt	member	memory
1113	1000 1011	3 1 3 4	9	12		mention	menu	mercy	merge	merit	merry	mesh	message
1121	1000 1100	3 1 4 1	9	13		metal	method	middle	midnight	milk	million	mimic	mind
1129	1000 1101	3 1 4 2	9	14		minimum	minor	minute	miracle	mirror	misery	miss	mistake
1137	1000 1110	3 1 4 3	9	15		mix	mixed	mixture	mobile	model	modify	mom	moment
1145	1000 1111	3 1 4 4	9	16		monitor	monkey	monster	month	moon	moral	more	morning
1153	1001 0000	3 2 1 1	10	1		mosquito	mother	motion	motor	mountain	mouse	move	movie
1161	1001 0001	3 2 1 2	10	2		much	muffin	mule	multiply	muscle	museum	mushroom	music
1169	1001 0010	3 2 1 3	10	3		must	mutual	myself	mystery	myth	naive	name	napkin
1177	1001 0011	3 2 1 4	10	4		narrow	nasty	nation	nature	near	neck	need	negative
1185	1001 0100	3 2 2 1	10	5		neglect	neither	nephew	nerve	nest	net	network	neutral
1193	1001 0101	3 2 2 2	10	6		never	news	next	nice	night	noble	noise	nominee
1201	1001 0110	3 2 2 3	10	7		noodle	normal	north	nose	notable	note	nothing	notice
1209	1001 0111	3 2 2 4	10	8		novel	now	nuclear	number	nurse	nut	oak	obey
1217	1001 1000	3 2 3 1	10	9		object	oblige	obscure	observe	obtain	obvious	occur	ocean
1225	1001 1001	3 2 3 2	10	10		october	odor	off	offer	office	often	oil	okay
1233	1001 1010	3 2 3 3	10	11		old	olive	olympic	omit	once	one	onion	online
1241	1001 1011	3 2 3 4	10	12		only	open	opera	opinion	oppose	option	orange	orbit
1249	1001 1100	3 2 4 1	10	13		orchard	order	ordinary	organ	orient	original	orphan	ostrich
1257	1001 1101	3 2 4 2	10	14		other	outdoor	outer	output	outside	oval	oven	over
1265	1001 1110	3 2 4 3	10	15		own	owner	oxygen	oyster	ozone	pact	paddle	page
1273	1001 1111	3 2 4 4	10	16		pair	palace	palm	panda	panel	panic	panther	paper
1281	1010 0000	3 3 1 1	11	1		parade	parent	park	parrot	party	pass	patch	path
1289	1010 0001	3 3 1 2	11	2		patient	patrol	pattern	pause	pave	payment	peace	peanut
1297	1010 0010	3 3 1 3	11	3		pear	peasant	pelican	pen	penalty	pencil	people	pepper
1305	1010 0011	3 3 1 4	11	4		perfect	permit	person	pet	phone	photo	phrase	physical
1313	1010 0100	3 3 2 1	11	5		piano	picnic	picture	piece	pig	pigeon	pill	pilot
1321	1010 0101	3 3 2 2	11	6		pink	pioneer	pipe	pistol	pitch	pizza	place	planet
1329	1010 0110	3 3 2 3	11	7		plastic	plate	play	please	pledge	pluck	plug	plunge
1337	1010 0111	3 3 2 4	11	8		poem	poet	point	polar	pole	police	pond	pony
1345	1010 1000	3 3 3 1	11	9		pool	popular	portion	position	possible	post	potato	pottery
1353	1010 1001	3 3 3 2	11	10		poverty	powder	power	practice	praise	predict	prefer	prepare
1361	1010 1010	3 3 3 3	11	11		present	pretty	prevent	price	pride	primary	print	priority
1369	1010 1011	3 3 3 4	11	12		prison	private	prize	problem	process	produce	profit	program
1377	1010 1100	3 3 4 1	11	13		project	promote	proof	property	prosper	protect	proud	provide
1385	1010 1101	3 3 4 2	11	14		public	pudding	pull	pulp	pulse	pumpkin	punch	pupil
1393	1010 1110	3 3 4 3	11	15		puppy	purchase	purity	purpose	purse	push	put	puzzle
1401	1010 1111	3 3 4 4	11	16		pyramid	quality	quantum	quarter	question	quick	quit	quiz
1409	1011 0000	3 4 1 1	12	1		quote	rabbit	raccoon	race	rack	radar	radio	rail
1417	1011 0001	3 4 1 2	12	2		rain	raise	rally	ramp	ranch	random	range	rapid
1425	1011 0010	3 4 1 3	12	3		rare	rate	rather	raven	raw	razor	ready	real
1433	1011 0011	3 4 1 4	12	4		reason	rebel	rebuild	recall	receive	recipe	record	recycle
1441	1011 0100	3 4 2 1	12	5		reduce	reflect	reform	refuse	region	regret	regular	reject
1449	1011 0101	3 4 2 2	12	6		relax	release	relief	rely	remain	remember	remind	remove
1457	1011 0110	3 4 2 3	12	7		render	renew	rent	reopen	repair	repeat	replace	report
1465	1011 0111	3 4 2 4	12	8		require	rescue	resemble	resist	resource	response	result	retire
1473	1011 1000	3 4 3 1	12	9		retreat	return	reunion	reveal	review	reward	rhythm	rib
1481	1011 1001	3 4 3 2	12	10		ribbon	rice	rich	ride	ridge	rifle	right	rigid
1489	1011 1010	3 4 3 3	12	11		ring	riot	ripple	risk	ritual	rival	river	road
1497	1011 1011	3 4 3 4	12	12		roast	robot	robust	rocket	romance	roof	rookie	room
1505	1011 1100	3 4 4 1	12	13		rose	rotate	rough	round	route	royal	rubber	rude
1513	1011 1101	3 4 4 2	12	14		rug	rule	run	runway	rural	sad	saddle	sadness
1521	1011 1110	3 4 4 3	12	15		safe	sail	salad	salmon	salon	salt	salute	same
1529	1011 1111	3 4 4 4	12	16		sample	sand	satisfy	satoshi	sauce	sausage	save	say

Lookup: words 1537 to 2048

D6 first roll 4, D20 first roll 13 to 16, coins begin 11. Find the row by your first entries and the column by your last.

					Coins: last 3 flips	000	001	010	011	100	101	110	111
					D6: 5th roll, 6th roll	1 1-2	1 3-4	2 1-2	2 3-4	3 1-2	3 3-4	4 1-2	4 3-4
					D20: 3rd roll	1-2	3-4	5-6	7-8	9-10	11-12	13-14	15-16
No.	**Coins**		**D6**		**D20**								
1537	1100	0000	4 1 1 1	13	1	scale	scan	scare	scatter	scene	scheme	school	science
1545	1100	0001	4 1 1 2	13	2	scissors	scorpion	scout	scrap	screen	script	scrub	sea
1553	1100	0010	4 1 1 3	13	3	search	season	seat	second	secret	section	security	seed
1561	1100	0011	4 1 1 4	13	4	seek	segment	select	sell	seminar	senior	sense	sentence
1569	1100	0100	4 1 2 1	13	5	series	service	session	settle	setup	seven	shadow	shaft
1577	1100	0101	4 1 2 2	13	6	shallow	share	shed	shell	sheriff	shield	shift	shine
1585	1100	0110	4 1 2 3	13	7	ship	shiver	shock	shoe	shoot	shop	short	shoulder
1593	1100	0111	4 1 2 4	13	8	shove	shrimp	shrug	shuffle	shy	sibling	sick	side
1601	1100	1000	4 1 3 1	13	9	siege	sight	sign	silent	silk	silly	silver	similar
1609	1100	1001	4 1 3 2	13	10	simple	since	sing	siren	sister	situate	six	size
1617	1100	1010	4 1 3 3	13	11	skate	sketch	ski	skill	skin	skirt	skull	slab
1625	1100	1011	4 1 3 4	13	12	slam	sleep	slender	slice	slide	slight	slim	slogan
1633	1100	1100	4 1 4 1	13	13	slot	slow	slush	small	smart	smile	smoke	smooth
1641	1100	1101	4 1 4 2	13	14	snack	snake	snap	sniff	snow	soap	soccer	social
1649	1100	1110	4 1 4 3	13	15	sock	soda	soft	solar	soldier	solid	solution	solve
1657	1100	1111	4 1 4 4	13	16	someone	song	soon	sorry	sort	soul	sound	soup
1665	1101	0000	4 2 1 1	14	1	source	south	space	spare	spatial	spawn	speak	special
1673	1101	0001	4 2 1 2	14	2	speed	spell	spend	sphere	spice	spider	spike	spin
1681	1101	0010	4 2 1 3	14	3	spirit	split	spoil	sponsor	spoon	sport	spot	spray
1689	1101	0011	4 2 1 4	14	4	spread	spring	spy	square	squeeze	squirrel	stable	stadium
1697	1101	0100	4 2 2 1	14	5	staff	stage	stairs	stamp	stand	start	state	stay
1705	1101	0101	4 2 2 2	14	6	steak	steel	stem	step	stereo	stick	still	sting
1713	1101	0110	4 2 2 3	14	7	stock	stomach	stone	stool	story	stove	strategy	street
1721	1101	0111	4 2 2 4	14	8	strike	strong	struggle	student	stuff	stumble	style	subject
1729	1101	1000	4 2 3 1	14	9	submit	subway	success	such	sudden	suffer	sugar	suggest
1737	1101	1001	4 2 3 2	14	10	suit	summer	sun	sunny	sunset	super	supply	supreme
1745	1101	1010	4 2 3 3	14	11	sure	surface	surge	surprise	surround	survey	suspect	sustain
1753	1101	1011	4 2 3 4	14	12	swallow	swamp	swap	swarm	swear	sweet	swift	swim
1761	1101	1100	4 2 4 1	14	13	swing	switch	sword	symbol	symptom	syrup	system	table
1769	1101	1101	4 2 4 2	14	14	tackle	tag	tail	talent	talk	tank	tape	target
1777	1101	1110	4 2 4 3	14	15	task	taste	tattoo	taxi	teach	team	tell	ten
1785	1101	1111	4 2 4 4	14	16	tenant	tennis	tent	term	test	text	thank	that
1793	1110	0000	4 3 1 1	15	1	theme	then	theory	there	they	thing	this	thought
1801	1110	0001	4 3 1 2	15	2	three	thrive	throw	thumb	thunder	ticket	tide	tiger
1809	1110	0010	4 3 1 3	15	3	tilt	timber	time	tiny	tip	tired	tissue	title
1817	1110	0011	4 3 1 4	15	4	toast	tobacco	today	toddler	toe	together	toilet	token
1825	1110	0100	4 3 2 1	15	5	tomato	tomorrow	tone	tongue	tonight	tool	tooth	top
1833	1110	0101	4 3 2 2	15	6	topic	topple	torch	tornado	tortoise	toss	total	tourist
1841	1110	0110	4 3 2 3	15	7	toward	tower	town	toy	track	trade	traffic	tragic
1849	1110	0111	4 3 2 4	15	8	train	transfer	trap	trash	travel	tray	treat	tree
1857	1110	1000	4 3 3 1	15	9	trend	trial	tribe	trick	trigger	trim	trip	trophy
1865	1110	1001	4 3 3 2	15	10	trouble	truck	true	truly	trumpet	trust	truth	try
1873	1110	1010	4 3 3 3	15	11	tube	tuition	tumble	tuna	tunnel	turkey	turn	turtle
1881	1110	1011	4 3 3 4	15	12	twelve	twenty	twice	twin	twist	two	type	typical
1889	1110	1100	4 3 4 1	15	13	ugly	umbrella	unable	unaware	uncle	uncover	under	undo
1897	1110	1101	4 3 4 2	15	14	unfair	unfold	unhappy	uniform	unique	unit	universe	unknown
1905	1110	1110	4 3 4 3	15	15	unlock	until	unusual	unveil	update	upgrade	uphold	upon
1913	1110	1111	4 3 4 4	15	16	upper	upset	urban	urge	usage	use	used	useful
1921	1111	0000	4 4 1 1	16	1	useless	usual	utility	vacant	vacuum	vague	valid	valley
1929	1111	0001	4 4 1 2	16	2	valve	van	vanish	vapor	various	vast	vault	vehicle
1937	1111	0010	4 4 1 3	16	3	velvet	vendor	venture	venue	verb	verify	version	very
1945	1111	0011	4 4 1 4	16	4	vessel	veteran	viable	vibrant	vicious	victory	video	view
1953	1111	0100	4 4 2 1	16	5	village	vintage	violin	virtual	virus	visa	visit	visual
1961	1111	0101	4 4 2 2	16	6	vital	vivid	vocal	voice	void	volcano	volume	vote
1969	1111	0110	4 4 2 3	16	7	voyage	wage	wagon	wait	walk	wall	walnut	want
1977	1111	0111	4 4 2 4	16	8	warfare	warm	warrior	wash	wasp	waste	water	wave
1985	1111	1000	4 4 3 1	16	9	way	wealth	weapon	wear	weasel	weather	web	wedding
1993	1111	1001	4 4 3 2	16	10	weekend	weird	welcome	west	wet	whale	what	wheat
2001	1111	1010	4 4 3 3	16	11	wheel	when	where	whip	whisper	wide	width	wife
2009	1111	1011	4 4 3 4	16	12	wild	will	win	window	wine	wing	wink	winner
2017	1111	1100	4 4 4 1	16	13	winter	wire	wisdom	wise	wish	witness	wolf	woman
2025	1111	1101	4 4 4 2	16	14	wonder	wood	wool	word	work	world	worry	worth
2033	1111	1110	4 4 4 3	16	15	wrap	wreck	wrestle	wrist	write	wrong	yard	year
2041	1111	1111	4 4 4 4	16	16	yellow	you	young	youth	zebra	zero	zone	zoo